

Cloud-Based System with GUI for Audio Classification Using Spectrograms and CNNs (GoogLeNet & AlexNet)

1st Noor S. Ali

Department of Electrical Engineering
College of Engineering
University of Babylon
Babylon, Iraq
eng.noor.salman@uobabylon.edu.iq

2nd Ehab Abdulrazzaq Hussein

Department of Electrical Engineering
College of Engineering
University of Babylon
Babylon, Iraq
dr.ehab@uobabylon.edu.iq

3rd Laith Ali Abdul-Rahaim

Department of Electrical Engineering,
College of Engineering
University of Babylon
Babylon Iraq
drlaithanzy@uobabylon.edu.iq

Abstract— Audio signals have a significant impact on every element of human life. The development of automated systems for various domains depends heavily on sound. There is an urgent need for cloud computing-based audio signal excellence systems, and their efficiency is a point of interest when implemented in real-life scenarios. Audio classification systems can be developed to solve the efficiency problems of conventional methods using the learning capabilities of cloud computing and deep learning architectures. This research proposes deep learning techniques to classify environmental audio signals based on the spectrograms generated for these sounds using cloud computing architecture. Spectrogram images of environmental audio signals train deep-learning networks such as GoogLeNet and AlexNet on cloud computing. Both networks are trained on a large number of datasets. The achieved accuracy can reach 92.5% on GoogLeNet and 87.5% on AlexNet. This paper provides that the proposed technique for audio signal classification using cloud computing can be efficient for sound classification and recognition systems.

Keywords — Audio Classification, Cloud Computing, Deep learning, GoogLeNet, AlexNet.

I. INTRODUCTION

The information sound carries about its source and the environment in which it was produced is rich. Humans can quickly process this complex sound and having a system capable of processing that equally quick is exactly useful, specially in scenarios where you have loud sounds being played [1]. In practice, sound classification is a critical front-end to many real-world and entertainment applications like emotion recognition, accent detection, speech-music separation and music retrieval [2]. Such operations, typically based on deep learning and CNNs rely on lots of computational calculation [3]. In this context, Many IoT systems have moved to the cloud, hence cloud-based processing is also used to meet them (new demands). The burgeoning cloud servers brought powerful remote capabilities for complex workloads, such as audio classification [4], image processing [5], or natural language translation [6] that alleviate the load on local devices now. Yet, this results in a very centralized system, where most of the computational processing occurs in the cloud [6].

Sound classification based on cloud computing is a large-scale classification that may be employed by innovative applications [7]. The Google dataset of sound classification can be selected for training deep learning networks on cloud computing. This implementation uses many systems to connect to the cloud and perform basic sound classification tasks [8]. The

audio is then classified according to particular patterns within the dataset, and DNN has been trained on it. Every device must transmit its audio data, which could expand the DNN dataset and improve the system's classification accuracy [9]. In the absence of cloud computing, the user device's application would require a longer runtime due to additional operations, perhaps leading to unjustifiable wait times for categorization. This wait time could be decreased by employing cloud computing and guaranteeing the integrity of incoming data, providing real-time audio representation in the region the system is categorizing [10]. By simulating the sound classification system, we develop a feasible and deployable real-time cloud-based voice classification system for sound classification.

This paper investigated the simplification and validation of DNN-based audio categorization and a transfer learning model that creates a deployable real-time audio classification system using cloud computing. Single and sequential representations have been subjected to image-based phonological representations (spectrograms and histograms), and the effects of these variations have been assessed. The study also offers some options for figuring out various transfer learning process parameters and looks into how they affect classification accuracy when taking particular values into account. It was confirmed that the algorithms and the input data were "interoperable." Compared to the accuracy attained using the classical approaches, the classification accuracy is higher. The classification accuracy was improved by the late fusion algorithms, which were based on the data obtained from various DNNs. Spectrograms were selected as input features, as they maintain the temporal and spectral properties of the audio signals and can successfully express the subtle audio differences. We use CNNs because they are powerful in dealing with two dimensional spectrograms and extracting discriminative patterns, which suits well with the accurate cloud-based audio classification.

II. RELATED WORK

Large-scale audio classification has gradually been moving to cloud computing. For example, in [11], Baucas discussed the difficulties of scaling for the massive amount of connected devices emphasizing the importance of edge computing as well the necessity of having deployable, real-time speech categorization systems with deep learning.

Baucas [12] later explored IoT networks with cloud and fog computing, showing via the case of urban sound detection that fog architectures reduced end-device power consumption and server latency, and offering a scalable solution for the application.

Meanwhile, Li [13] explored various feature extraction methods (e.g., the Binaural MFCC feature, the log Mel-spectrum feature, large-scale pooling feature) and machine learning models (GMM, DNN, RNN, and CNN) for comparison. The model outperforms baseline by 11% in test accuracy and 15.6% in cross-validation accuracy, suggesting the benefit of using advanced features for deep learning.

In the spirit of our prior work [19], we designed a framework based on GoogLeNet and AlexNet that could classify environmental audio signals with 92.5% and 87.5% accuracy, respectively. It did not have a cloud contract and GUI. In this study, we develop upon that work by establishing a cloud-based platform with a built-in GUI that makes the technology more user-friendly and permits real-time analysis.

In general, previous works aimed at increasing classification accuracy with different neural network models and either cloud-based or fog-based infrastructure. We are motivated to improve on these works and hence, introduce a cloud-based system with GUI this paper that can classify audio using spectrograms and transfer learning for CNNs (GoogLeNet and AlexNet). This approach not only takes advantage of large-scale audio corpora, but also can keep up in real time processing in a cloud computing setting.

TABLE I. SUMMARY OF RELATED WORKS

Ref	Method	Dataset	Accuracy	Contribution
[11]	Edge/Fog computing for speech categorization	IoT data	—	Proposed scalable framework
[12]	Fog computing for IoT sound detection	IoT networks	—	Reduced latency, better power efficiency
[13]	MFCC, log Mel + GMM/DNN/RNN/CNN	OpenSMILE features	+11% improvement in test accuracy	Improved baseline with hybrid features

This work	Spectrograms + GoogLeNet/AlexNet	Google Audio Dataset	High accuracy	Cloud-based GUI, real-time classification
------------------	----------------------------------	----------------------	---------------	---

As shown in TABLE I, unlike previous studies that relied mainly on MFCC or fog computing frameworks, our proposed approach leverages spectrograms with CNN architectures (GoogLeNet and AlexNet) in a cloud environment to achieve real-time classification with high accuracy

III. CLOUD COMPUTING FRAMEWORK

Cloud computing offers access to a shared pool of computing resources on the fly for large scale audio classification [14], [15]. In this paper, cloud infrastructure is utilized to assist real-time classification and reduce computational burden to the devices. The complete IoT–cloud architecture is shown in Fig. 1

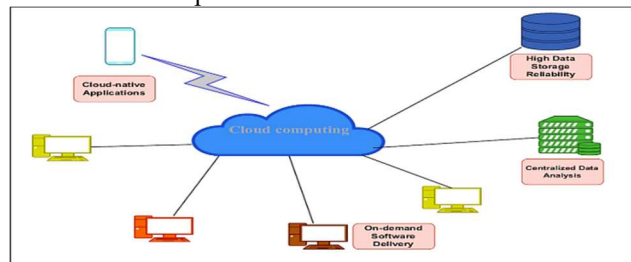


Fig. 1. IoT Cloud network.

A. DATASET AND PREPROCESSING

Experiments were also conducted using the Google AudioSet dataset [16]. All of the recordings were resampled to 16 kHz, normalized, and then cut into 2–4 seconds of sound clips. Preprocess additionally included framing and filtering for enhanced features extraction (Fig. 2).

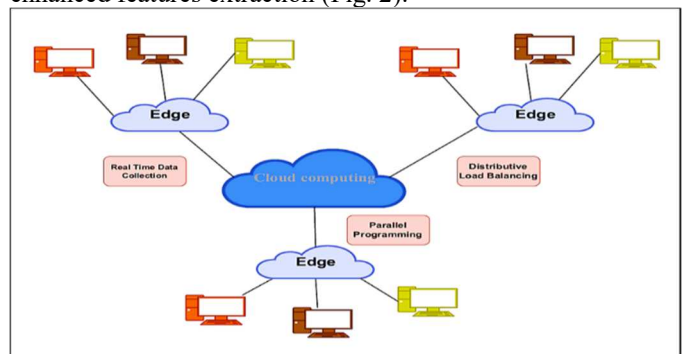


Fig. 2. Edge-based IoT network

B. SPECTROGRAM FEATURE EXTRACTION

For each audio clip, it was applied the Short-Time Fourier Transform (STFT). Parameters: Hamming window, 50% incumbent, 512 FFT points [17]. The produced 2D spectrograms retain spectral as well as temporal content and were used as the input of the CNN models.

C. CNN MODELS: GOOGLNET AND ALEXNET.

Two transfer learning models were adopted: AlexNet (8 layers, 5 conv layers followed by 3 fully connected layers), good for small and medium size data as well. GoogLeNet[18]: A deeper network with Inception modules that enables multi-scale feature extraction. The last fully connected layer of both models were replaced to accommodate the number of target classes.

D. TRAINING SETUP

Experiments were conducted in MATLAB R2023b and we utilized GPU for fast training. Two optimizers were compared: SGDM: batch size = 15, epochs = 25, learning rate = $1e-4$. Adam: batch size of 64, 30 epochs, learning rate of $1e-4$. Performance measures were Accuracy, Precision, Recall, and F1-score.

E. GUI AND PROPOSED SYSTEM

A graphical user interface (GUI) was created to enable the user to upload audio files, create spectrograms, and classify them with trained CNNs. Two configurations were tested: Edge-based: classification done on-device (Fig. 3).

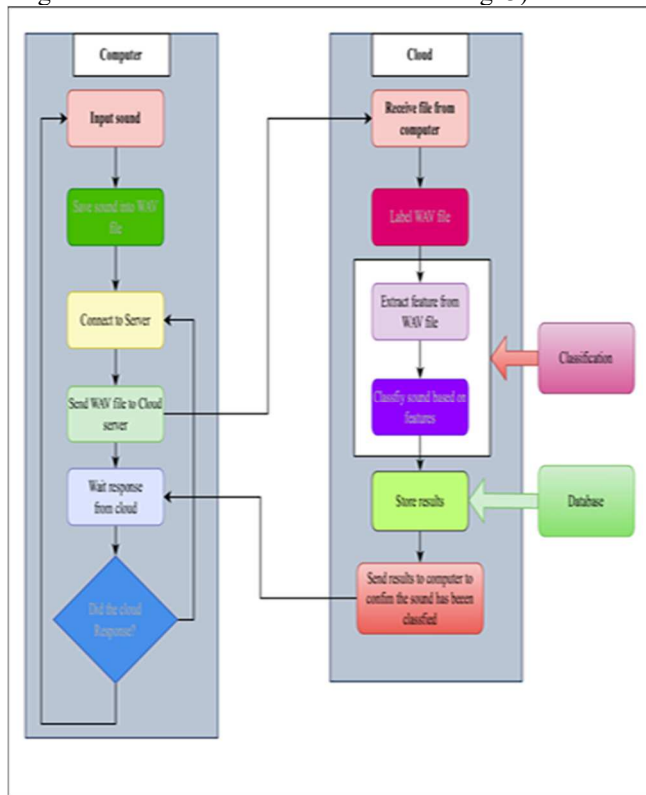


Fig. 3. Cloud-based proposed design.

Audio is sent to the server for analysis with Spectrogram and Classification happening in the Cloud (Fig. 4).

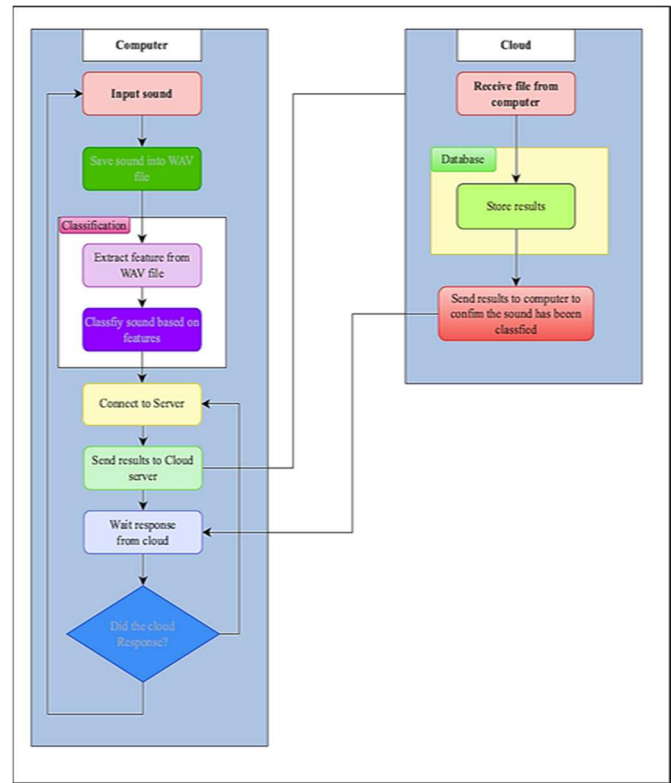


Fig. 4. Edge Cloud-based proposed design.

Energy and latency were compared to analyze scalability of both designs.

VI. RESULTS

A script that records audio via the microphone has been put into place. After the recording, the software saves the audio clip as a WAV file. An additional script that reads the WAV file and extracts its features comes next. These features are subsequently used by the audio classifier model described in the neural network section. Upon receiving the model's results through cloud computing. Four main sections make up the experimental setup. Every component supports the previously covered data processing methods and their application as a reliable classifier. The following sections are included: Testbed Setup, Sound Classifier, Feature extraction, and MATLAB Design. The final step included the analysis of what is, in computing terms, known as regular-performance tests. The configurations differ as follows: the first configuration considers edge classification, audio files processed locally and stored in the server; the second configuration contemplates cloud classification, where audio files are transmitted into the server, reconstructed, classified and finally stored into a

database.

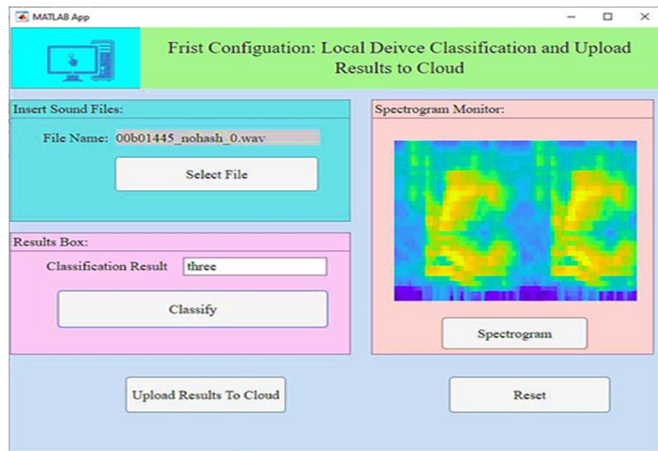


Fig.5. The computer GUI of the first configuration

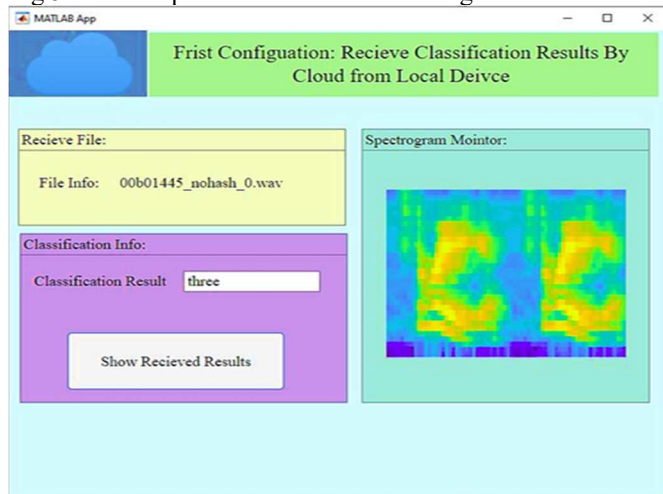


Fig .6. The Cloud GUI of the first configuration.

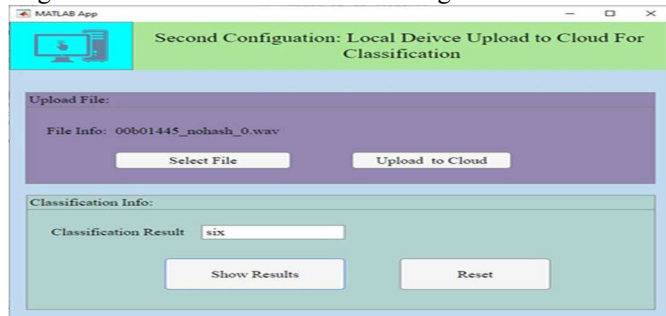


Fig.7.The computer GUI of the Second configuration.

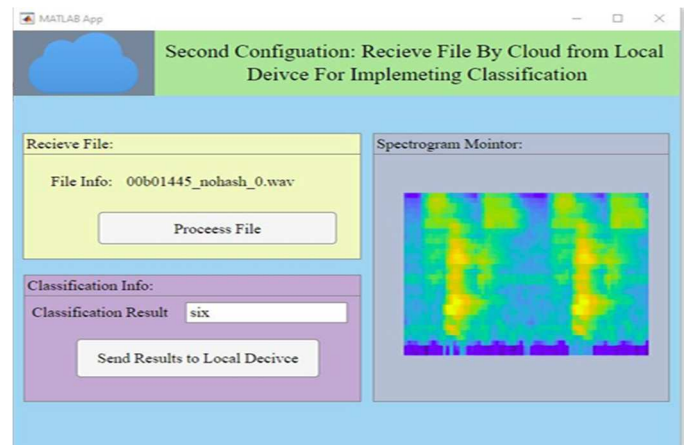


Fig.8. The Cloud GUI of the Second configuration.

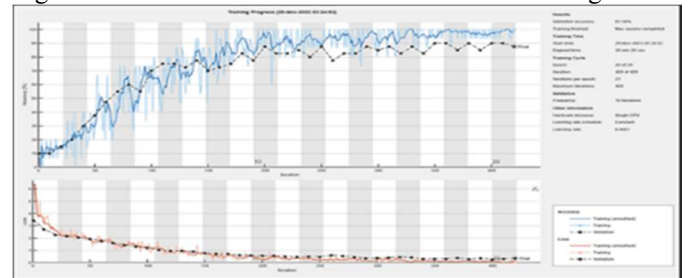


Fig.9. The Training Progress of AlexNet with SGDM
The training accuracy curves for AlexNet with SGDM are depicted in Fig. 9, showing the convergence trends of model

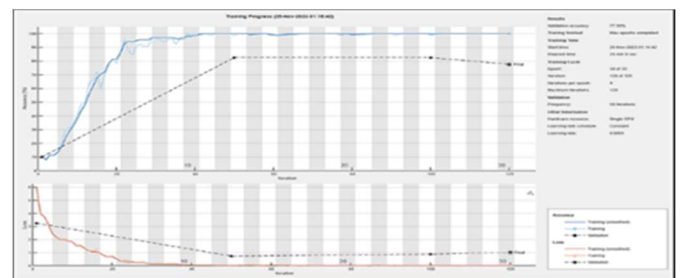


Fig.10. The Training Progress of AlexNet with ADAM.
The training accuracy curve for AlexNet with ADAM is shown in Fig. 9, which presents the convergence behaviors of model

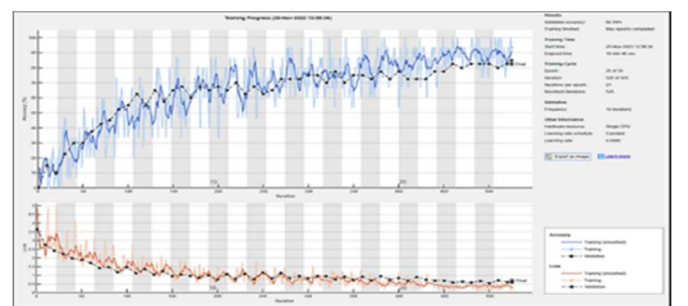


Fig.11. The Training Progress of GoogleNet with SGDM.

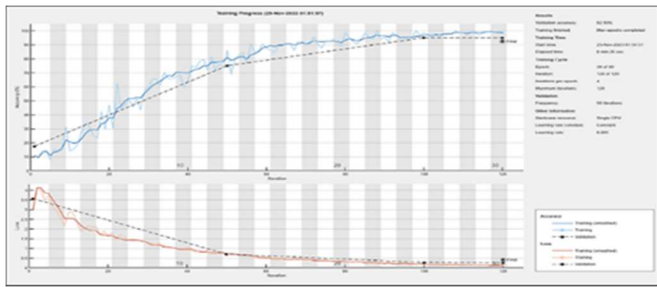


Fig.12. The Training Progress of GoogleNet with ADAM.

While SGDM outperformed the Adam optimizer in terms of training time, the ADAM optimizer outperformed it in terms of accuracy. Furthermore, as anticipated, the training time decreased with increasing mini-batch size and reversed for the number of epochs. A highly sensitive hyperparameter to adjust is the learning rate, which determines how quickly and well a network learns. Compared to SGDM, ADAM outperformed it in every Image CNN. Table 4 compares the networks based on the average values of training time and Accuracy for every optimizer in the dataset. The network with the lowest Accuracy and the highest training time was AlexNet. GoogleNet displayed the shortest training time, which was almost 2.5 times higher than AlexNet's equivalent. Table II. Accuracy and training time for each network for both optimizers.

CNN	Adam		SGDM	
	CA (%)	TT (s)	CA (%)	TT (s)
AlexNet	77.5	1509	87.5	1820
GoogleNet	92.5	648	82.5	505

The Image CNNs were always trained with the Adam optimizer, a batch size of 64 and a learning rate of 1×10^{-4} . The number of epochs were based on different dataset so we might find it looks similar due to the difference in the network architecture and diversity of data. For the test sound group, classification results were obtained from all best performing configurations specified before with one pseudo-randomly chosen test file per configuration. Spectrograms and scalograms for the same audio data, are trained (80/20 validation split) Time (sec) Dataset 21466 Table 10 : Pre-processing time for Scalogram generation across the dataset. Figures 13–16 illustrate confusion matrices.

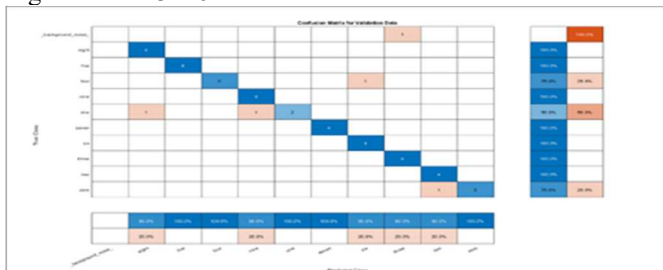


Fig.13.The confusion matrix of AlexNet with SGDM.

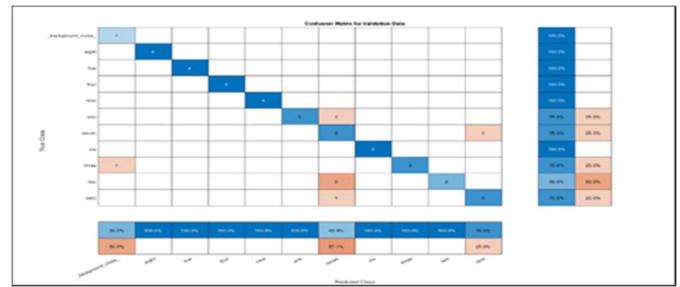


Fig.14. The confusion matrix of AlexNet with ADAM.

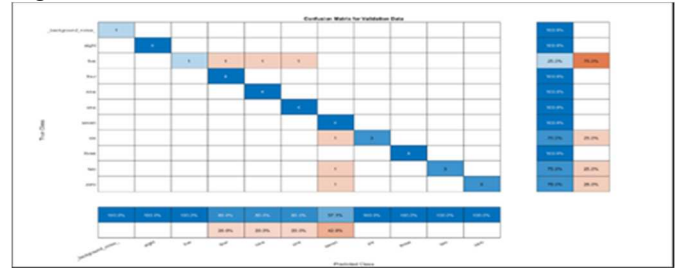


Fig.15 The confusion matrix of GoogleNet with SGDM.

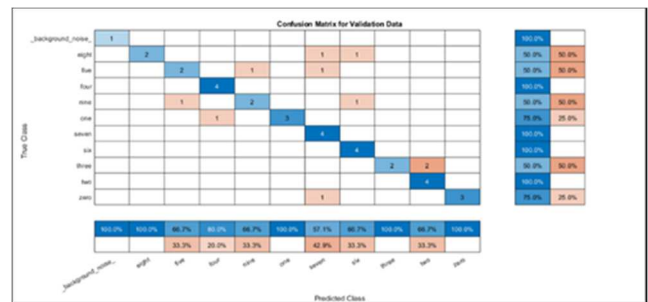


Fig.16. The confusion matrix of GoogleNet with ADAM.

V.CONCLUSION

Edge computing and cloud computing are the two forms of computing that are the subject of this study. Every kind concentrates on a different facet of the Internet of Things. Experiments were carried out to explore optimization strategies for developing a system that minimizes server-side strain and end-device processing for audio classification. The classifier was implemented within the end device in the first setup. In the second arrangement, the classifier was relocated to the server. First, the classifier's performance was evaluated. We tracked the system's runtime and processing consumption for each setting. The second configuration was better due to the server's processing power than the end device. Regarding audio representation, comparable image-based transformations (such as spectrograms and scalograms) have produced similar outcomes when preferences and limitations are beforehand. Positive results are also possible in merger circumstances. Audio CNNs are one method of achieving fusion with a single classifier. Other methods include audio segmentation, separate classifications for each audio, or merging the results. Fusion may also happen if several classifiers are available for a majority-based classification. The two CNNs worked better together on the Google speech dataset than separately. In both CNN merging scenarios, 92.5% classification accuracy was attained. Comparative feedback on the quantity and quality of the dataset, such as sound segmentation, labeling accuracy, and

species clarity (or the overlapping of numerous sounds), is made possible by the classification accuracy of the same CNNs on various datasets. Zero exhibited the best behavior for Google's voice datasets since it provided the highest classification accuracy among all CNNs. This collection has many audio files, which are organized into eleven categories.

REFERENCES

- [1] H. Abdullah et al., "Practical hidden voice attacks against speech and speaker recognition systems," arXiv preprint arXiv:1904.05734, 2019.
- [2] A. E. Youssef, "Exploring cloud computing services and applications," *Journal of Emerging Trends in Computing and Information Sciences*, vol. 3, no. 6, pp. 838–847, 2012.
- [3] J. Chen and X. Ran, "Deep learning with edge computing: A review," *Proceedings of the IEEE*, vol. 107, no. 8, pp. 1655–1674, 2019.
- [4] F. Jauaro et al., "Deep learning architectures in emerging cloud computing architectures: Recent development, challenges, and next research trend," *Applied Soft Computing*, vol. 96, p. 106582, 2020.
- [5] Q. Qi and F. Tao, "A smart manufacturing service system based on edge computing, fog computing, and cloud computing," *IEEE Access*, vol. 7, pp. 86769–86777, 2019.
- [6] S. K. Shah, Z. Tariq, and Y. Lee, "IoT based urban noise monitoring in deep learning using historical reports," in *Proc. IEEE Int. Conf. on Big Data (Big Data)*, 2019, pp. 5940–5942.
- [7] J.-W. Jung et al., "DNN-based audio scene classification for DCASE2017: Dual input features, balancing cost, and stochastic data duplication," in *Proc. DCASE Workshop*, 2017.
- [8] M. Mohaimenuzzaman, C. Bergmeir, and B. Meyer, "Pruning vs XNOR-net: A comprehensive study of deep learning for audio classification on edge-devices," *IEEE Access*, vol. 10, pp. 6696–6707, 2022.
- [9] M. J. Baucas and P. Spachos, "Using cloud and fog computing for large scale IoT-based urban sound classification," *Simulation Modelling Practice and Theory*, vol. 101, p. 102013, 2020.
- [10] M. J. Baucas and P. Spachos, "Using cloud and fog computing for large scale IoT-based urban sound classification," arXiv preprint arXiv:1910.05679, 2019.
- [11] I. Odun-Ayo et al., "An overview of data storage in cloud computing," in *Proc. Int. Conf. on Next Generation Computing and Information Systems (ICNGCIS)*, 2017, pp. 29–34.
- [12] I. Zhou et al., "Internet of things 2.0: Concepts, applications, and future directions," *IEEE Access*, vol. 9, pp. 70961–71012, 2021.
- [13] X. Zhang et al., "A survey on privacy inference attacks and defenses in cloud-based deep neural network," *Computer Standards & Interfaces*, vol. 83, p. 103672, 2023.
- [14] D. T. Várkonyi, J. L. S. Junior, and T. Horváth, "Dynamic noise filtering for multi-class classification of beehive audio data," *Expert Systems with Applications*, vol. 213, p. 118850, 2023.
- [15] H. Elazhary, "Internet of Things (IoT), mobile cloud, cloudlet, mobile IoT, IoT cloud, fog, mobile edge, and edge emerging computing paradigms: Disambiguation and research directions," *Journal of Network and Computer Applications*, vol. 128, pp. 105–140, 2019.
- [16] Y. Wu, "Cloud-edge orchestration for the Internet of Things: Architecture and AI-powered data processing," *IEEE Internet of Things Journal*, vol. 8, no. 16, pp. 12792–12805, 2020.
- [17] A. A. Alvarez, *Deep Learning Methods for Motivic Pattern Extraction and Classification*, Ph.D. dissertation, ETSIS Telecomunicación, 2021.
- [18] S. Minaee et al., "Biometrics recognition using deep learning: A survey," *Artificial Intelligence Review*, pp. 1–49, 2023.
- [19] N. S. Ali, E. A. Hussein, and L. A. Abdul-Rahaim, "Classification of audio signals based on deep learning methods for GoogLeNet and AlexNet," in *Proc. 2024 3rd International Conference on Advances in Engineering Science and Technology (AEST)**, Babil, Iraq, Nov. 24–25, 2024. doi: 10.1109/AEST63017.2024.10960116.